

Lab 2 – Firewall Protecting Virtual Machines

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Marks

Objective

To understand how Virtualization can be used to protect Virtual Machines connected to a Virtual Network.

Background

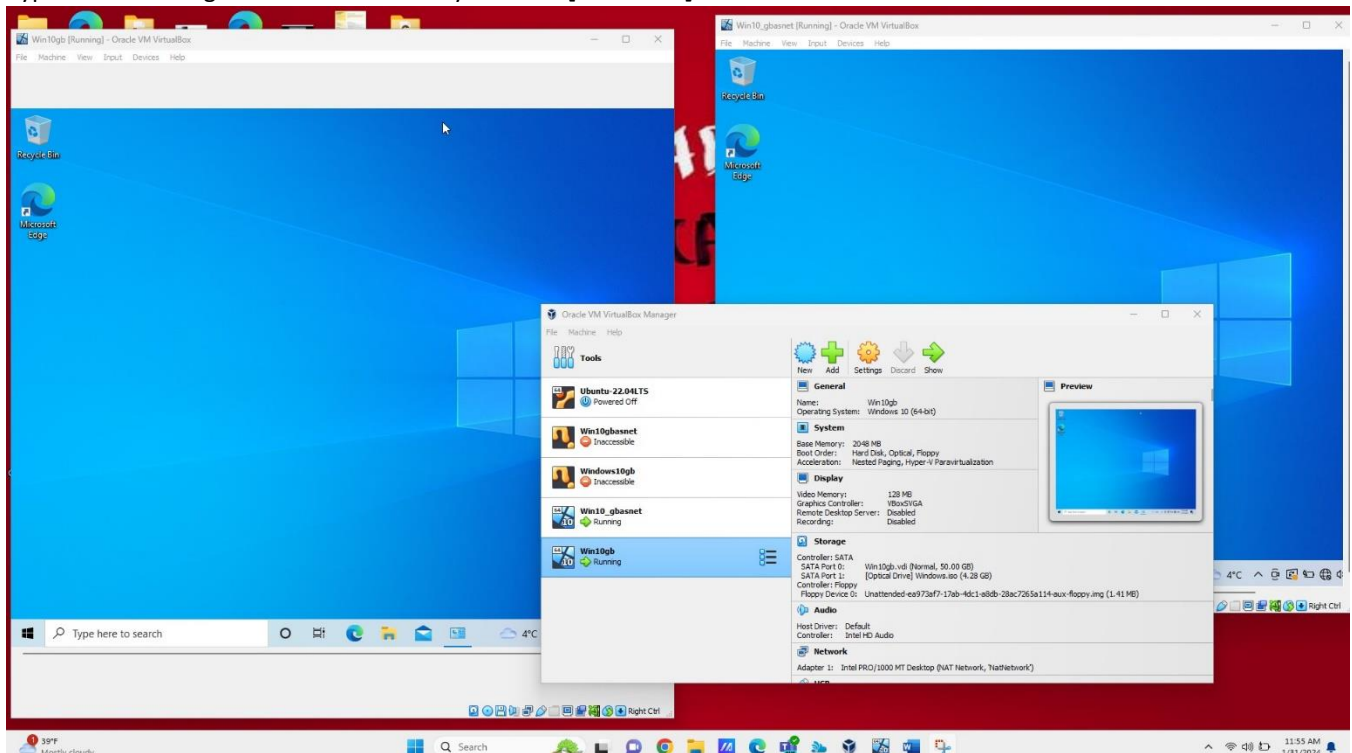
- Use the material posted from week 2&3 to verify the Network Functions Virtualization concepts.
- In previous courses you have learned about Network Security Protocols. You can also explore the firewall settings on Windows or Linux. This lab shows examples of Windows 10 VMs and Firewall settings on Windows 10.

Instructions

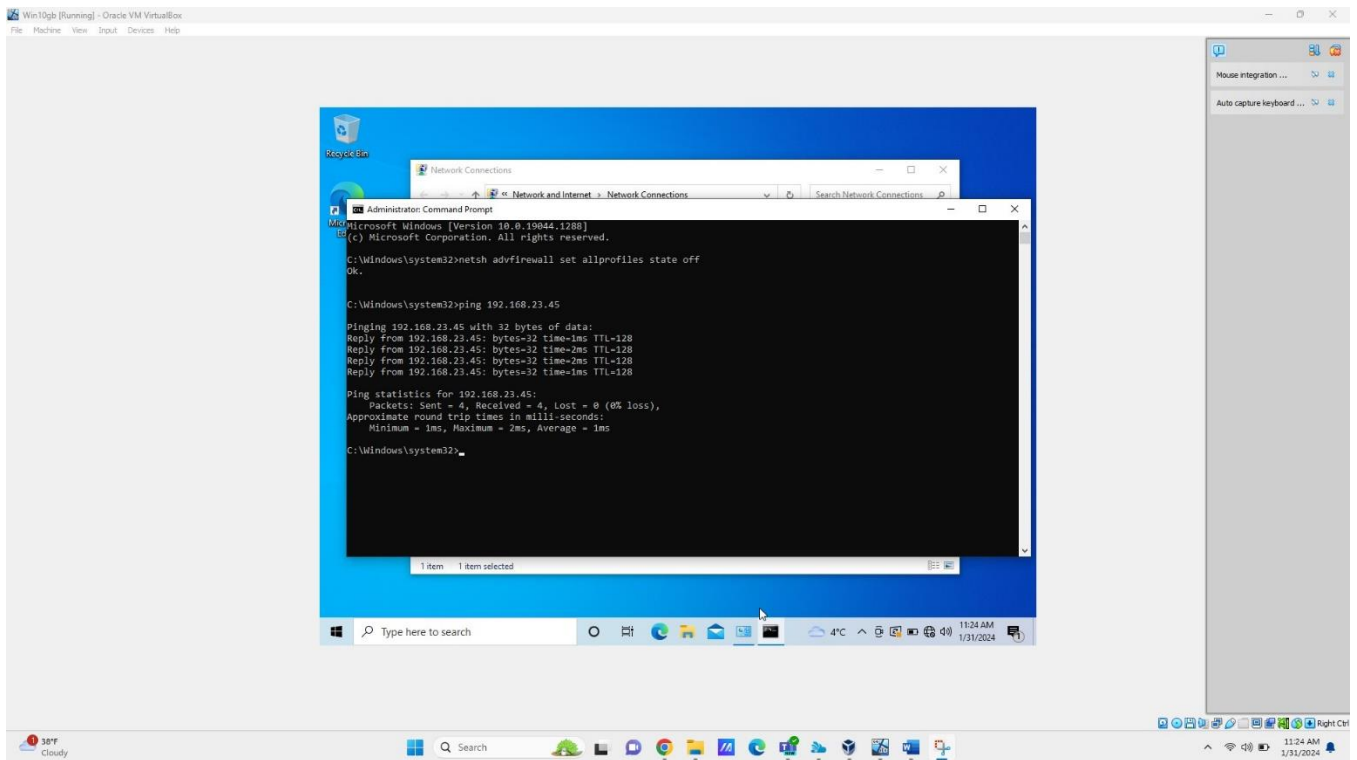
1. Use a different font and color to write your answers.
2. Do not erase anything from this document. Insert your answers and screenshots without erasing or modifying the format of this document. Your screenshots must show your name and the date/time from your own host machine.
3. Resize the screenshots to make them easily readable.

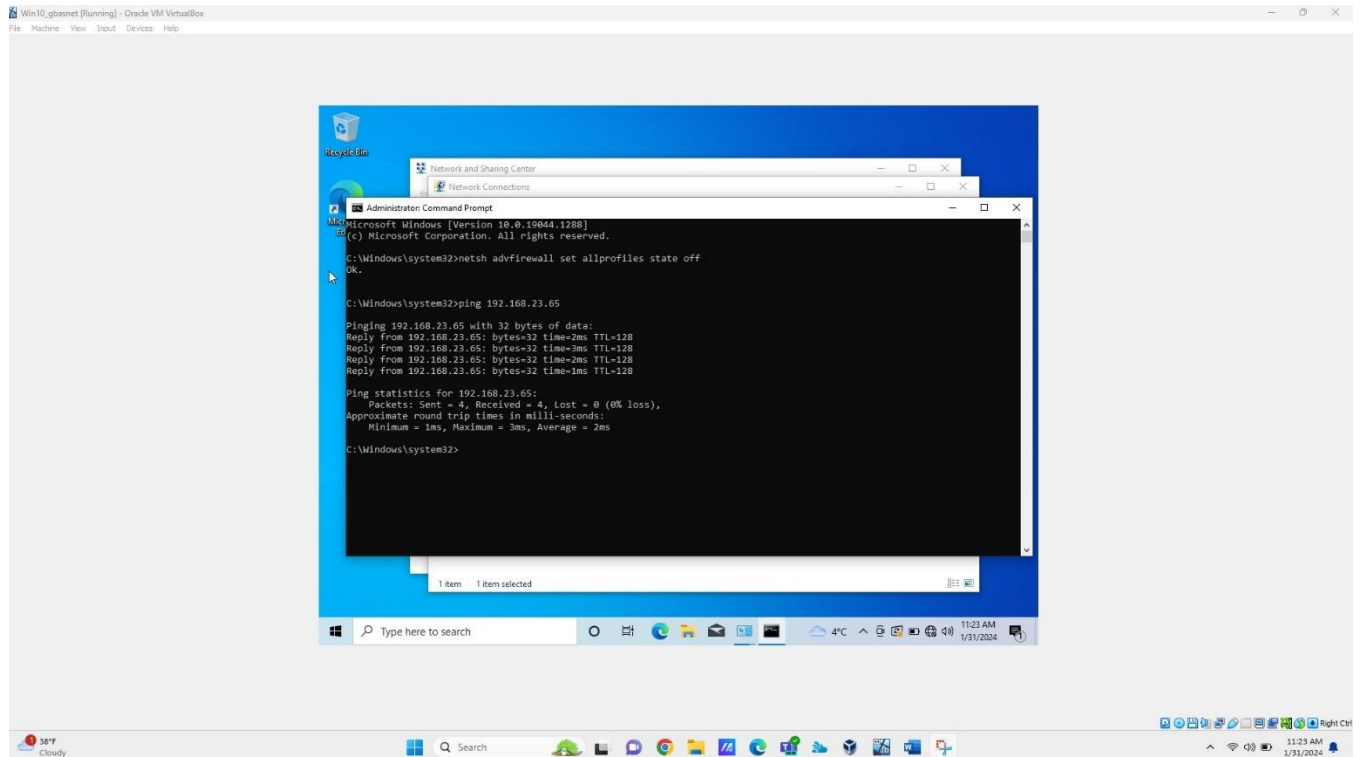
Procedure

1. Create two Windows 10 VMs (you can use previously created VMs if they are available). Take a screenshot of your hypervisor showing the list of VMs already created. [One mark]



2. Define a virtual network where the two VMs will be connected. Assign the following static addresses: (replace the xx digits by the two last digits of your student ID number)
 - VM1 IP address: 192.168.23.45
 - VM2 IP address: 192.168.23.65
3. What will be the gateway address? ____192.168.23.1____ [One mark].
4. Be sure that you properly set the two VMs to belong to this network. Check connectivity between them. Insert the screenshot of successful pings VM1_to_VM2 here. [One mark]

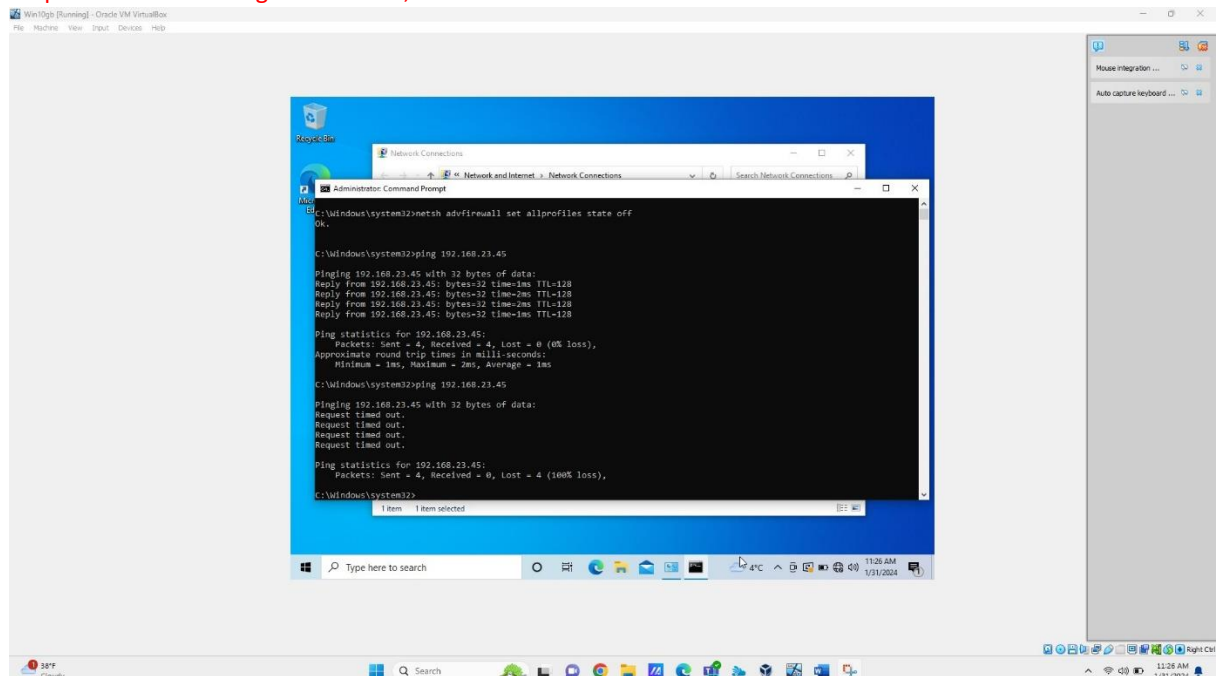




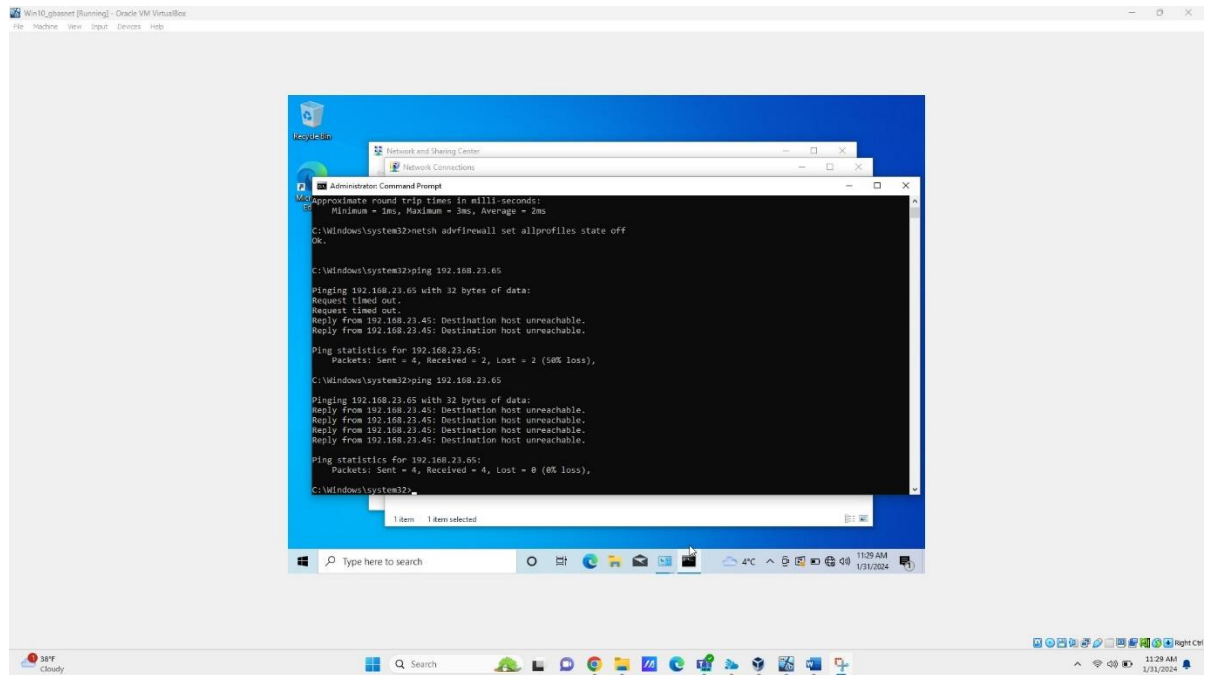
5. Now, modify the network configurations to get ping results as follows:
 - a. Request timed out
 - b. Destination host unreachable
 - c. General Failure

Explain what you do to create each of them (what is your configuration) and take screenshot to show the result for each one. **[Three marks]**

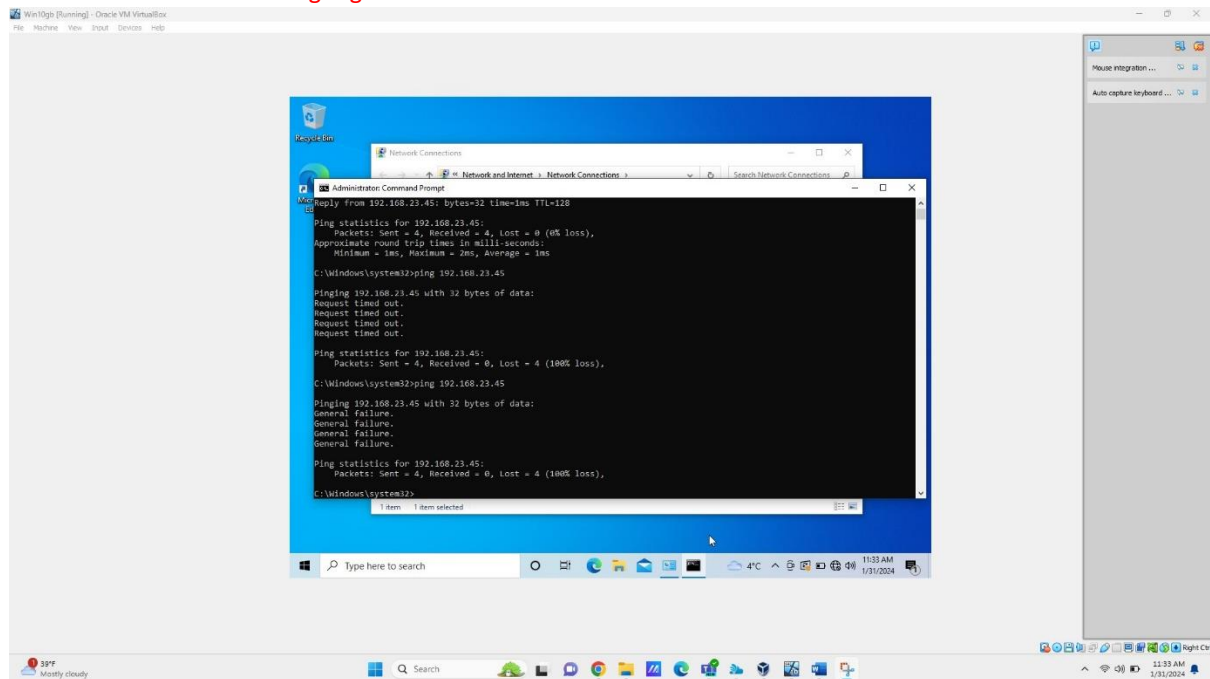
- a. **Request timed out:** To get this result, first we should enable the machine's firewall.



- b. **Destination Host Unreachable:** In order to get this first we need to disable the network adapter on the machine



- c. **General Failure:** In order to get general failure we must set inbound rule on machine to block ICMP.

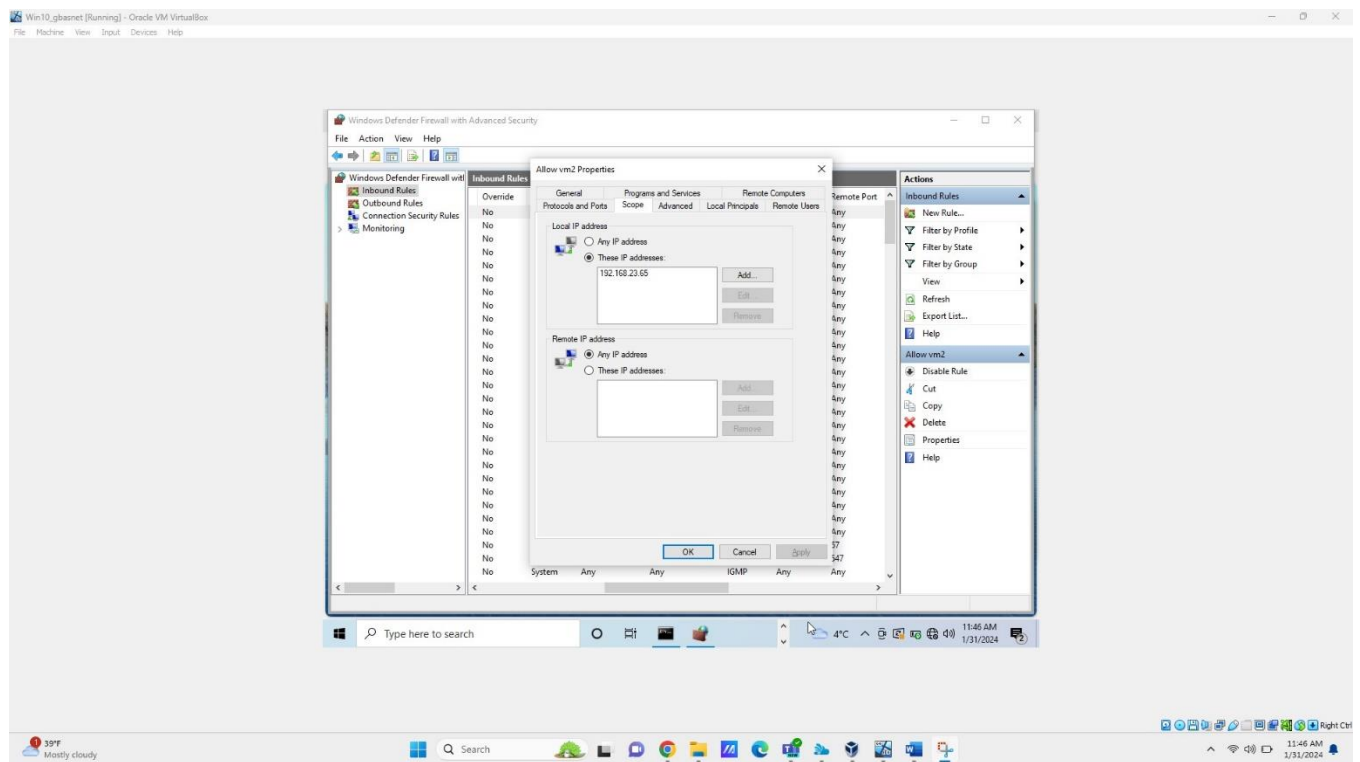


- Fix the network settings again to have a successful ping between VM1 and VM2. Now, visit the Microsoft's website about [how to configure the Windows 10 firewall](#).
- Activate the Windows Firewall in both VMs. Create a new Inbound rule in firewall in VM2, try to accept PING commands from VM1 to VM2. Be sure that you use the correct TCP or UDP port. Explain its scope and other

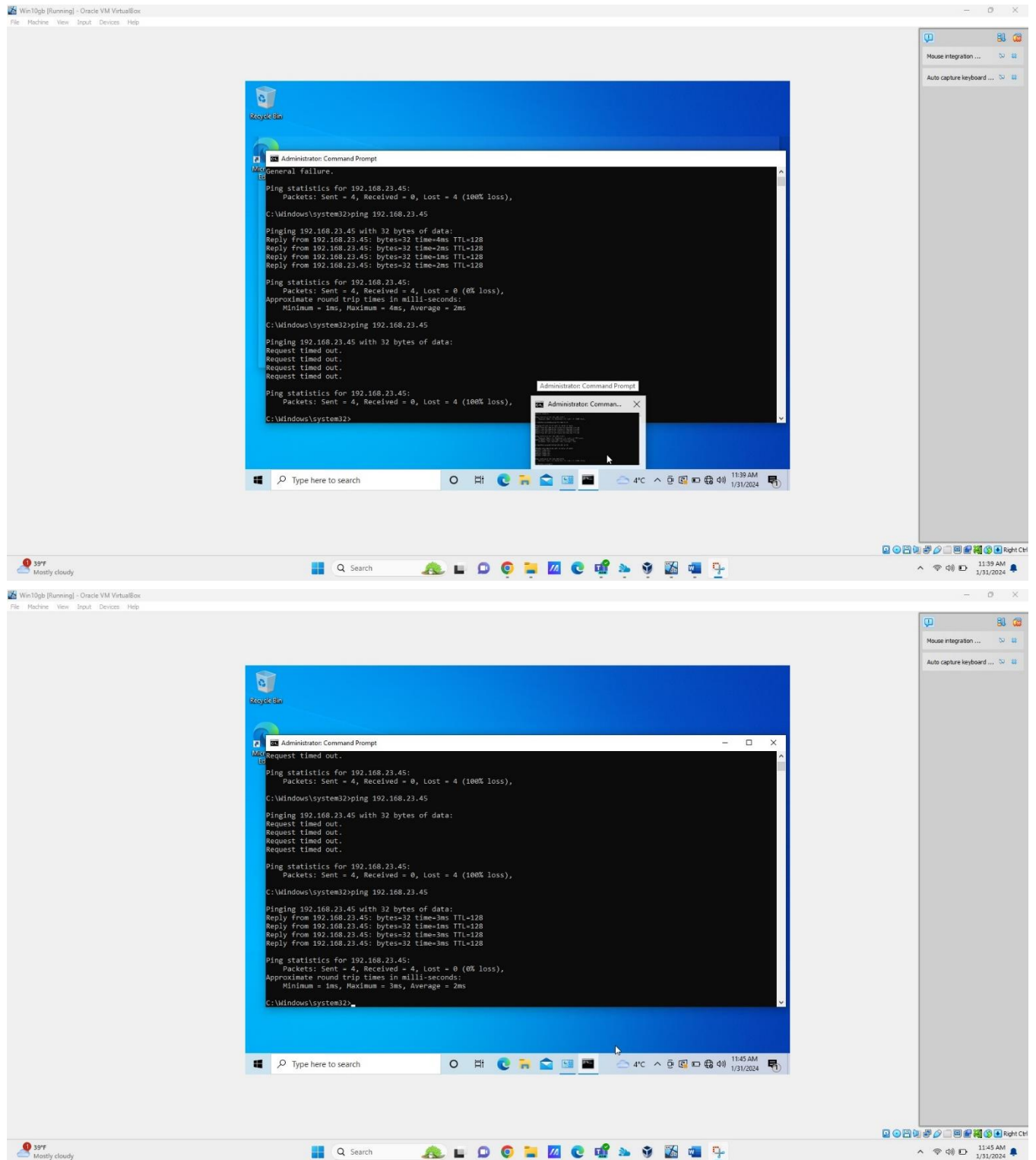
features of the rule if necessary. Explain how you created the rule and what are your configurations and take a screenshot from the completed rule. Explain why this rule can help PING to be successful. **[One mark]**

Answer: We must configure an inbound rule in VM2's firewall for it to accept PING commands from VM1 to VM2. Navigate to the Windows firewall settings to create this rule. Next, choose the new rule by clicking on the incoming tab. Establish the virtual machine's rule to allow ping requests. This regulation permits inbound ICMP allowing successful pinging and facilitating communication between VM1 and VM2.

Take a screenshot of the VM2 Firewall newly created rule configuration. Insert it here. **[One mark]**



Take two screenshots from PING to show that it does not work before this rule, and it works after creating this new rule. **[Two marks]**



To submit: (three files to be submitted)

1. Remember to work individually. Avoid plagiarism or any other type of cheating offense.
2. Be sure that you wrote your full name at the beginning of this document. (10% deducted if you do not write your name)
3. Rename this file as **LastNameFirstName_CSN305Lab2** and save it as PDF. Upload this document including all your answers and screenshots. (10% deducted if this file is missed in your submission or wrong name is used)
4. Submit the **VirtualBox Machine Definition** XML file corresponding to each VM you used in this lab. Submit these two files along with the report. (10% deducted for each missed VM definition/configuration file)